Planned: space–time 3D CNN (cnn3d_st): one ignition map per hour of the 12 h period

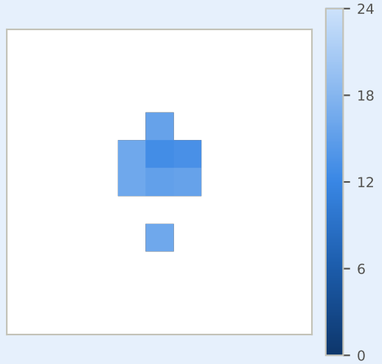
Same encoder as cnn3d_final. New: a decoder brings the time axis back to hourly steps, so the output keeps the hour of each ignition instead of pooling time away.

1 Input x

24 hourly steps × 7 channels
(frp, confidence, detected, period,
time, pt mask, persistent)
shape 24 × 7 × H × W

6 h history · 12 h period (blue) · 6 h tail

example: 2016-08-20 06–18 UTC (validation),
11 × 11 cells; colour = hour of first detection
(darker = earlier), SEVIRI only



2 Encoder: compress time

E1 Conv3d 3×3×3

7 → 16 channels, ReLU

24 steps of 1 h

skip

E2 Conv3d, stride 2 in t

16 → 32 channels, ReLU

12 steps of 2 h

E3 Conv3d, stride 2 in t

32 → 32 channels, ReLU

6 steps of 4 h

3 Decoder: restore hours (new)

D1 upsample + Conv3d

↑ to 24 steps, joined with E1
→ 16 channels

24 steps of 1 h

skip

D2 upsample + Conv3d

↑ to 12 steps, joined with E2
→ 32 channels

12 steps of 2 h

Global context (as in cnn3d_final)

max and mean of E3 over its 6 steps → 64 channels, copied to every hour:
each hour also 'knows' the whole 24 h window (did it keep burning? was it there before?)

Keep the period

hours 7–18 of D1
→ 16 × 12 × H × W

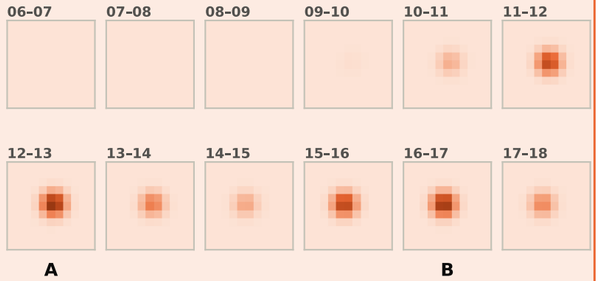
4 Head, every hour

join 16 + 64 channels
Conv3d 1×3×3 → 32, ReLU
Conv 1×1 → 1 logit
sigmoid → p(hour, i, j)

≈ 140k parameters (now 63k)

5 Output: 12 hourly maps

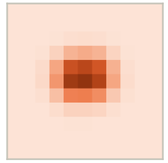
shown: target y (Gaussian σ = 1 cell, σ_t = 1 h)



A starts 6.2 h, B 10.2 h into the period (UTC hours shown);
0.8 cells apart: two separate peaks in (hour, i, j)

decode: 3D local maxima (3 h × 3 × 3 cells)
→ points (lat, lon, hour, score)

Compared with cnn3d_final



now: time is pooled away →
one 2D map: A and B merge
into one bump and one peak,
so one of them is missed

new: same encoder + decoder + per-hour head;
every ignition also gets an hour.
≈ 2.3× training time per step